

Mugisha Sangwa Olivier Ishimwe

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Education

Worcester Polytechnic Institute

B.S.; M.S., Computer Science, Artificial Intelligence

• **Coursework:** Data Structures, Algorithms, Operating Systems, Object-Oriented Programming, Functional Programming, Introduction to Artificial Intelligence & Machine Learning, Introduction to Cybersecurity

Jan 2026 - May 2028

Worcester, MA

Husson University

B.S., Software Development, Computer Information Systems

Aug 2024 - Dec 2025

Bangor, ME

Technical Skills

- **Languages:** Java, Python, C++, C#, JavaScript, TypeScript, Kotlin, Swift, SQL, HTML, CSS
- **Frameworks & Libraries:** React, Node.js, FastAPI, Flask, Capacitor, REST APIs
- **Platforms & Tools:** Git, Docker, Google Cloud Platform, PostgreSQL, Supabase, Stripe, VS Code, IntelliJ, Visual Studio
- **Concepts:** Data Structures & Algorithms, Object-Oriented & Functional Programming, System Design, Operating Systems, Testing

Projects

eMed | Founder & Lead Engineer - eMed

Jan 2026 - Present

- Founded and engineered a HIPAA-conscious healthcare-access platform that connects patients with physicians for on-demand virtual consultations, helping users avoid costly ER visits for non-emergency care.
- Shipped native iOS, Android, and web clients from a single TypeScript codebase using Capacitor, integrating responsive HTML/CSS for the web UI and eliminating duplicated platform-specific code.
- Designed RESTful backend services in Python with FastAPI on a normalized PostgreSQL schema, then containerized with Docker and deployed to Google Cloud Platform for reproducible builds across environments.

eNotary | Founder & Lead Engineer - eNotary

Jan 2026 - Present

- Founded and built a digital notarization platform for the Rwandan market that matches clients with available private notaries, reducing notarization turnaround from days to seconds.
- Engineered a real-time matching engine in Python to pair clients with qualified notaries by availability and document type, achieving low-latency responses under high concurrent demand.
- Built a secure document upload workflow with a React and JavaScript frontend (HTML/CSS) and a Python backend, ensuring tamper-evident audit trails for legal integrity and end-to-end traceability.

Smart MEP | Lead Software Engineer - Smart MEP

Jan 2026 - Present

- Lead engineering on a platform that automates the design and layout of mechanical, electrical, and plumbing (MEP) systems, replacing manual drafting for construction projects.
- Built automation logic that generates and validates electrical and plumbing layouts against design rules, substantially reducing drafting time and human error.
- Own technical direction and architecture decisions across a multi-language stack, driving core features from design through implementation, testing, and deployment.

Experience

Worcester Polytechnic Institute

Peer Learning Assistant - Computer Science Department

Worcester, MA • **Aug 2026 - Present**

- Lead weekly CS lab sessions using VS Code, Python, and Git, grade assignments, and provide targeted feedback, improving students' problem-solving and coding skills and boosting assignment grades

Program Coordinator - Pre-Collegiate Outreach Programs

Worcester, MA • **May 2026 - Present**

- Help ensure the smooth day-to-day running of WPI's pre-college programs by answering phone calls and emails, and help plan activities for student recess

Husson University

Teaching Assistant

Bangor, ME • **Sep 2025 - Dec 2025**

- Contributed to a 100% course pass rate by providing targeted one-on-one and small-group tutoring to reinforce complex data structures & algorithms concepts
- Facilitated technical discussions and clarified difficult material using IntelliJ and Node.js examples, then provided structured feedback that boosted students' analytical skills, leading to higher assignment scores

Student Assistant - Library

Bangor, ME • **May 2025 - Aug 2025**

- Built a full-stack data application using Flask and Docker that ingested and analyzed library circulation data to justify book-weeding decisions, enabling data-driven collection management
- Processed and visualized usage statistics to surface underutilized titles, reducing institutional costs by \$12,439 annually by identifying redundant academic subscriptions for cancellation.
- Optimized SQL queries against PostgreSQL to improve application performance by 30%